

Team Meeting

Oracle APEX · Endless Aisle

Titan / Tanishq UAT

Apex QA Agent Platform

Architecture, folder structure, plugin model, and delivery plan —
deterministic-first runtime with LLM for planning only.

Date: 4 September 2026 · **GT session:** today with SME · **Status:** local demo ready

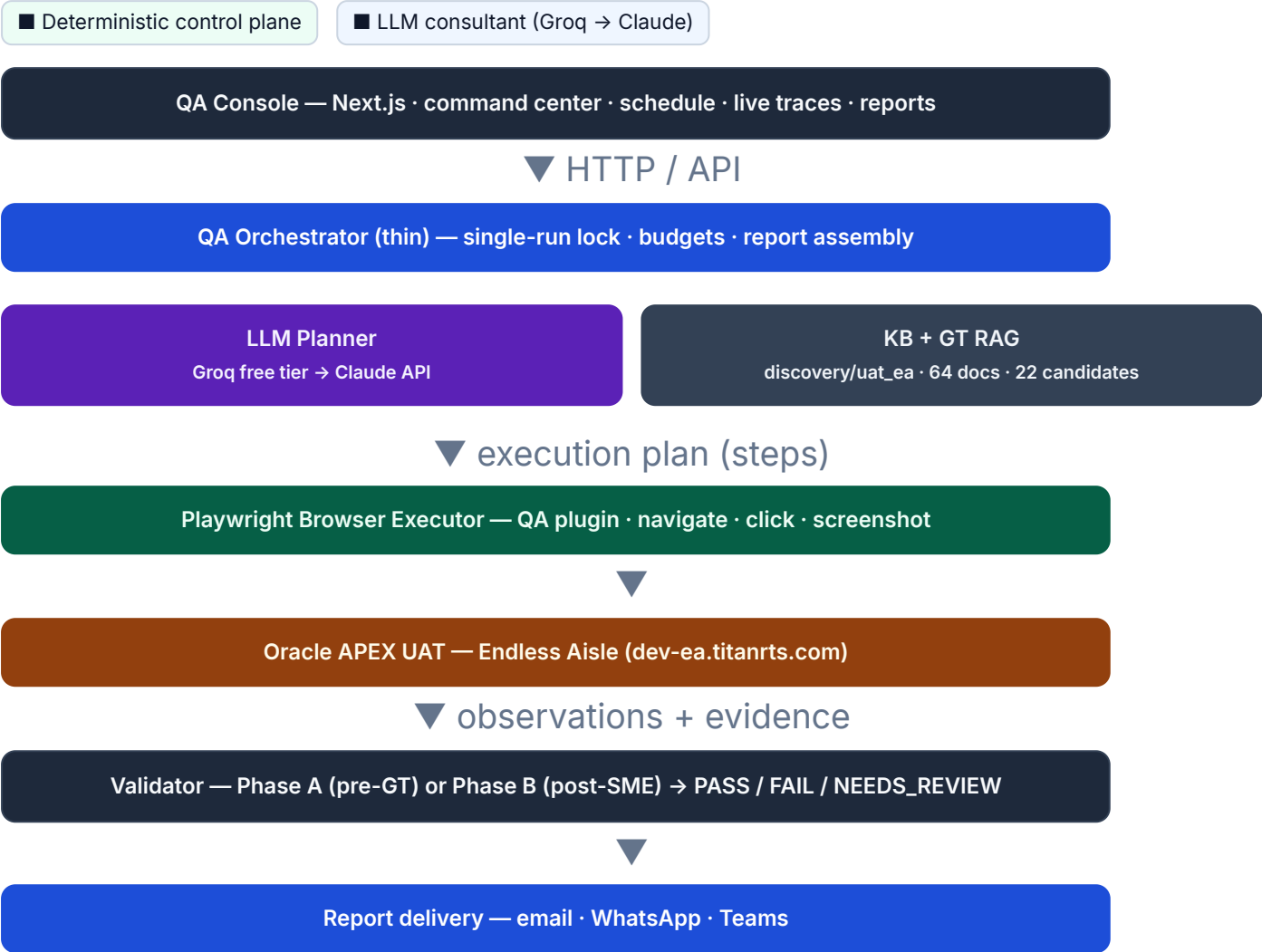


One-liner: Morning sanity + ad-hoc natural-language checks + multi-channel reports. One honest run per command — no loop-until-success. LLM plans; Playwright executes; Ground Truth validates after SME approval.

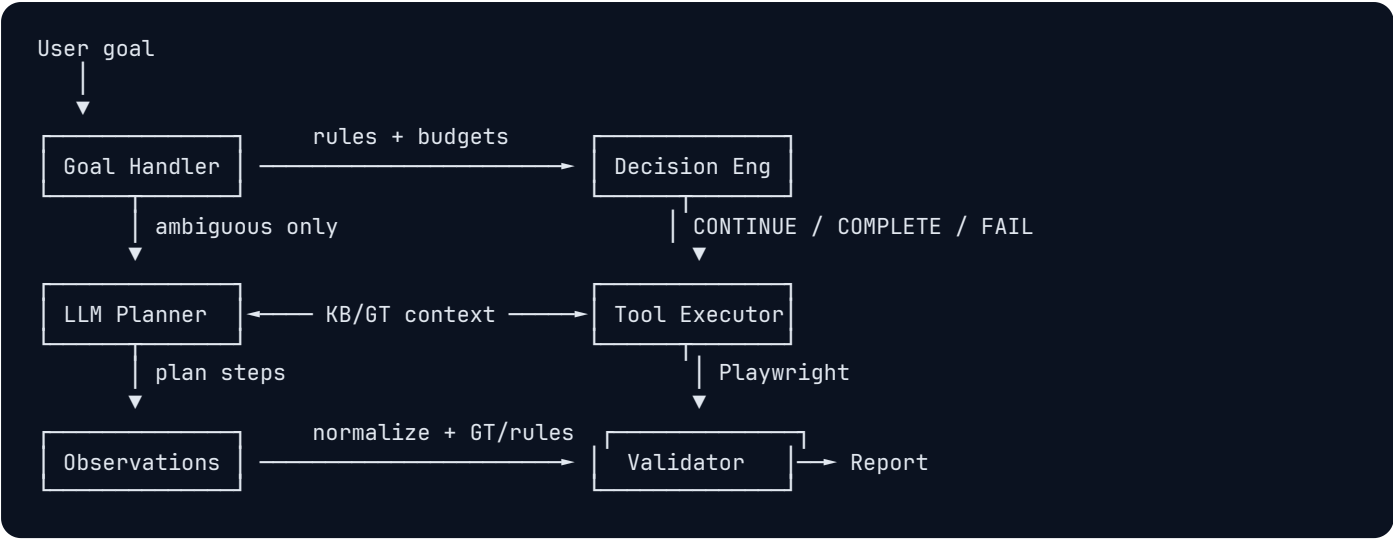
1. Client requirement → our scope

Client need	Platform answer
Morning sanity (scheduled)	08:00 IST patrol → login + home + key modules → report
Ad-hoc “check this”	NL command in console → LLM plan → Playwright run → report
Email / WhatsApp / Teams reports	Structured markdown + JSON evidence bundle
Works before Ground Truth	Phase A: technical rules + honest NEEDS_REVIEW
Stronger after SME GT (today)	Phase B: deterministic expected vs actual

2. System architecture (linear)



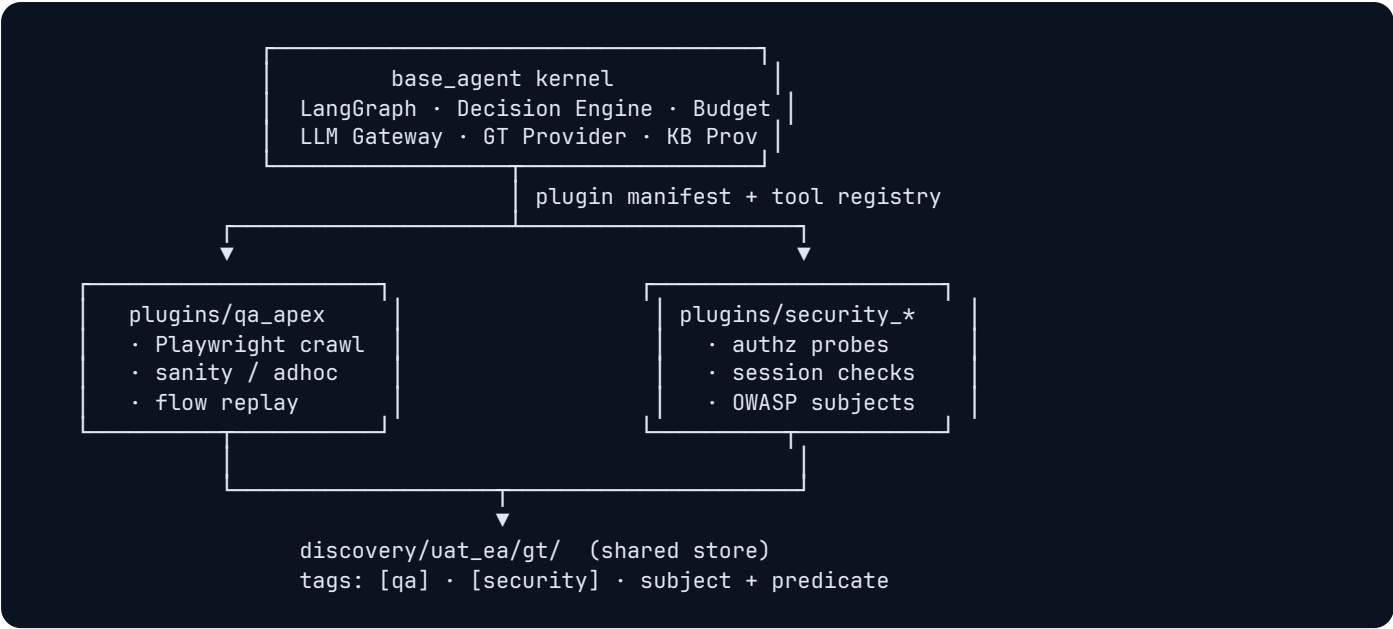
2.1 Control-plane flow (deterministic-first)



2.2 LLM vs deterministic split

Layer	Mode	Responsibility
Budgets, routing, run lock	Deterministic	max steps/tools/LLM — never loop-until-success
KB / GT lookup	Deterministic	RAG over JSON; GT compare after SME approval
NL goal → plan	LLM	Groq now; Claude API when client ready
Browser automation	Deterministic	Playwright steps, screenshots, timeouts
Technical checks	Deterministic	login dead, crash, timeout, stuck modal
Business checks (pre-GT)	LLM + rules	NEEDS_REVIEW — never fake PASS
Business checks (post-GT)	Deterministic	Approved facts → PASS / FAIL
Report summary	LLM (optional)	Short narrative for channels

3. Plugin model — QA + Security on one kernel



QA Agent (now): browser flows, morning patrol, ad-hoc NL commands, evidence reports.

Security Agent (later): same kernel and GT shapes — adds subjects like `authz.idor`, `session.fixation`. No runtime fork.

4. Ground Truth — SME session today

Asset	Count	Location
KB documents	64	discovery/uat_ea/kb/
GT candidates	22	discovery/uat_ea/candidate_gt/
Approval checklist	—	discovery/uat_ea/APPROVAL_CHECKLIST.md
Browser recordings merged	2 sessions	discovery/uat_ea/recordings/

Phase	When	Validation
A — Pre-GT	Now	Technical PASS/FAIL; business → NEEDS_REVIEW
B — Post-GT	After SME approves today	Deterministic expected vs actual

5. Repository folder structure (working tree)

Full project layout — console, runtime kernel, orchestrator, plugins, discovery pack, CI/CD.



- └─ page.tsx
- └─ components/
 - └─ ui/
 - └─ button.tsx
 - └─ input.tsx
 - └─ qa-console.tsx
- └─ data/
 - └─ state.json
- └─ lib/
 - └─ agent-runner.ts
 - └─ notify.ts
 - └─ store.ts
 - └─ types.ts
 - └─ utils.ts
- └─ public/
- └─ components.json
- └─ Dockerfile
- └─ eslint.config.mjs
- └─ next-env.d.ts
- └─ next.config.ts
- └─ package-lock.json
- └─ package.json
- └─ postcss.config.mjs
- └─ README.md
- └─ tsconfig.json
- └─ src/
 - └─ base_agent/
 - └─ budget/
 - └─ __init__.py
 - └─ guard.py
 - └─ capabilities/
 - └─ __init__.py
 - └─ context/
 - └─ __init__.py
 - └─ contracts/
 - └─ __init__.py
 - └─ enums.py
 - └─ models.py
 - └─ result.py
 - └─ decision/
 - └─ __init__.py
 - └─ engine.py
 - └─ errors/
 - └─ __init__.py
 - └─ taxonomy.py
 - └─ graph/
 - └─ nodes/
 - └─ __init__.py
 - └─ __init__.py
 - └─ builder.py
 - └─ ground_truth/
 - └─ __init__.py
 - └─ protocol.py
 - └─ knowledge/
 - └─ __init__.py
 - └─ protocol.py
 - └─ llm/
 - └─ __init__.py
 - └─ gateway.py
 - └─ observability/
 - └─ __init__.py
 - └─ observation/
 - └─ __init__.py
 - └─ pipeline.py
 - └─ plugins/
 - └─ __init__.py
 - └─ routing/
 - └─ __init__.py
 - └─ hybrid.py
 - └─ security/
 - └─ __init__.py
 - └─ skills/
 - └─ __init__.py
 - └─ state/
 - └─ __init__.py
 - └─ schema.py
 - └─ tools/
 - └─ __init__.py
 - └─ registry.py
 - └─ __init__.py
 - └─ api.py

- └─ base_agent.egg-info/
- └─ qa_orchestrator/
 - └─ __init__.py
 - └─ api.py
 - └─ kb_rag.py
 - └─ llm_client.py
 - └─ models.py
 - └─ browser_executor.py
 - └─ orchestrator.py
 - └─ planner.py
 - └─ reporter.py
 - └─ validator.py
- └─ plugins/
 - └─ mock_demo/
 - └─ __init__.py
 - └─ tools.py
 - └─ qa_apex/
 - └─ crawler/
 - └─ __init__.py
 - └─ apex_waits.py
 - └─ engine.py
 - └─ selectors.py
 - └─ session.py
 - └─ __init__.py
 - └─ rules.py
 - └─ skills.py
 - └─ tools.py
- └─ discovery/
 - └─ uat_ea/
 - └─ candidate_gt/
 - └─ candidates.json
 - └─ kb/
 - └─ app.endless_aisle.overview.json
 - └─ component.find_price.item.json
 - └─ component.find_price.lotno.json
 - └─ component.item_search.sku.json
- └─ component.product_detail.size_variants.json
- └─ component.standard_search.gpt.json
- └─ component.stock_visibility.sku.json
- └─ flow.best_deal_detail.json
- └─ flow.browse_all_products.json
- └─ flow.category_earrings.json
- └─ flow.find_price.json
- └─ flow.item_search.json
- └─ flow.item_sku_search.json
- └─ flow.login_home.json
- └─ flow.reports_menu.json
- └─ flow.rivaah_browse.json
- └─ flow.rivaah_trousseau.json
- └─ flow.standard_product_browse.json
- └─ flow.stock_visibility.json
- └─ index.json
- └─ nav.home_modules.json
- └─ page.administration.json
- └─ page.customer-wish-list1.json
- └─ page.dashboard-tile.json
- └─ page.discount-product-details-all-
- └─ store.json
- └─ page.dream-list-product1.json
- └─ page.eal.101.json
- └─ page.eal.102.json
- └─ page.eal.200.json
- └─ page.eal.22.json
- └─ page.eal.47.json
- └─ page.eal.51.json
- └─ page.eal.57.json
- └─ page.eal.61.json
- └─ page.eal.63.json
- └─ page.eal.estimation-slip.json
- └─ page.eal.grid-bills-test.json
- └─ page.eal.look-of-the-week-product-
- └─ detail.json
- └─ page.eal.manual-bills-book.json
- └─ page.eal.new-collections.json
- └─ page.eal.product-detail.json
- └─ page.eal.product-discount.json
- └─ page.eal.rakhi-collections-product-
- └─ details1.json
- └─ page.eal.silver-collections1.json
- └─ page.eal.size-based-search.json

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├── page.ea1.testing1.json
├── page.estimate-slip.json
├── page.find-price.json
├── page.gc.store-gold-coin-new.json
├── page.genai-agents-for-
apex105109125128.chat-with-oci-generative-ai-agent.json
├── page.gold-coin-stock-visibility.json
├── page.home.json
├── page.kvi-fmc-lcg.json
├── page.login.json
├── page.product-catalogue.json
├── page.product-detail-item-search.json
├── page.product-detail.json
├── page.product-discount.json
├── page.rivaah.json
├── page.select-category.json
├── page.smart-image-search1.json
├── page.solitaire-uin-search.json
├── page.standard-product-search.json
├── page.wedding-trousseau-details1.json
├── page.wedding-trousseau.json
├── pattern.apex.auth_home.json
├── pattern.apex.interactive_grid.json
├── pattern.apex.modal_dialog.json
├── pattern.apex.module_list_detail.json
├── pattern.apex.popup_lov.json
├── kb_normalized/
├── business/
├── │ app.endless_aisle.overview.json
├── components/
├── │ component.find_price.item.json
├── │ component.find_price.lotno.json
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├── index/
├── │ catalog.json
├── pages/
├── │ nav.home_modules.json
├── │ page.administration.json
├── │ page.ea.customer-wish-list1.json
├── │ page.ea.dashboard-tile.json
├── │ page.ea.discount-product-details-
all-store.json
├── │ page.ea.dream-list-product1.json
├── │ page.ea.estimate-slip.json
├── │ page.ea.find-price.json
├── │ page.ea.gold-coin-stock-
visibility.json
├── │ page.ea.home.json
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├── │ page.ea.login.json
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├── │ page.gc.store-gold-coin-new.json
├── │ page.genai-agents-for-
apex105109125128.chat-with-oci-generative-ai-agent.json
├── │ recordings/
├── │ │ merged_from_browser_recordings.json
├── │ │ report_58aa.md
├── │ │ report_8d0d.md
├── │ │ APPROVAL_CHECKLIST.md
├── │ │ discovery_report.json
├── │ │ kb_summary.json
├── scripts/
├── │ local_agent_server.py
├── │ morning_patrol.py
├── deploy/
├── │ Dockerfile
├── tests/
├── │ unit/
├── │ │ test_crawler_qa.py
├── │ │ test_qa_orchestrator.py
├── │ │ test_qa_skills.py
├── │ │ test_runtime.py
├── │ conftest.py
├── docs/
├── │ architecture/
├── │ │ APEX_APPLICATION_FLOWS.md
├── │ │ APEX_CRAWLER_PERFORMANCE.md
├── │ │ AZURE_OCI_DEPLOYMENT.md
├── │ │ LOCAL_RUN.md
├── │ │ OPERATING_WITHOUT_GT.md
├── │ │ QA_ORCHESTRATOR.md
├── │ │ REPORT_CHANNELS.md
├── │ │ SIMPLIFIED_QA_ARCHITECTURE.md
├── │ meeting/
├── │ │ screenshots/
├── │ │ TEAM_MEETING_ARCHITECTURE.md
├── │ APEX_GT_KB_COLLECTION_SPEC.md
├── │ BASE_AGENT_TECHNICAL_PROPOSAL.md
├── │ PHASE_COMPLETE.md
├── │ PRODUCT_STATUS.md
├── azure-pipelines.yml
├── pyproject.toml
├── README.md

```

6. Requirements to complete

QA Agent (priority)

Item	Owner
GROQ_API_KEY	Dev
ANTHROPIC_API_KEY (later)	Client/Dev
APEX_TARGET_URL	Client
APEX_USERNAME / APEX_PASSWORD	Client vault
SME GT approval (today)	SME
SMTP / SendGrid	Client IT
Twilio / WhatsApp Business	Client IT
Teams webhook	Client IT
Azure DevOps + service connections	Client IT
OCI registry + vault	Client IT

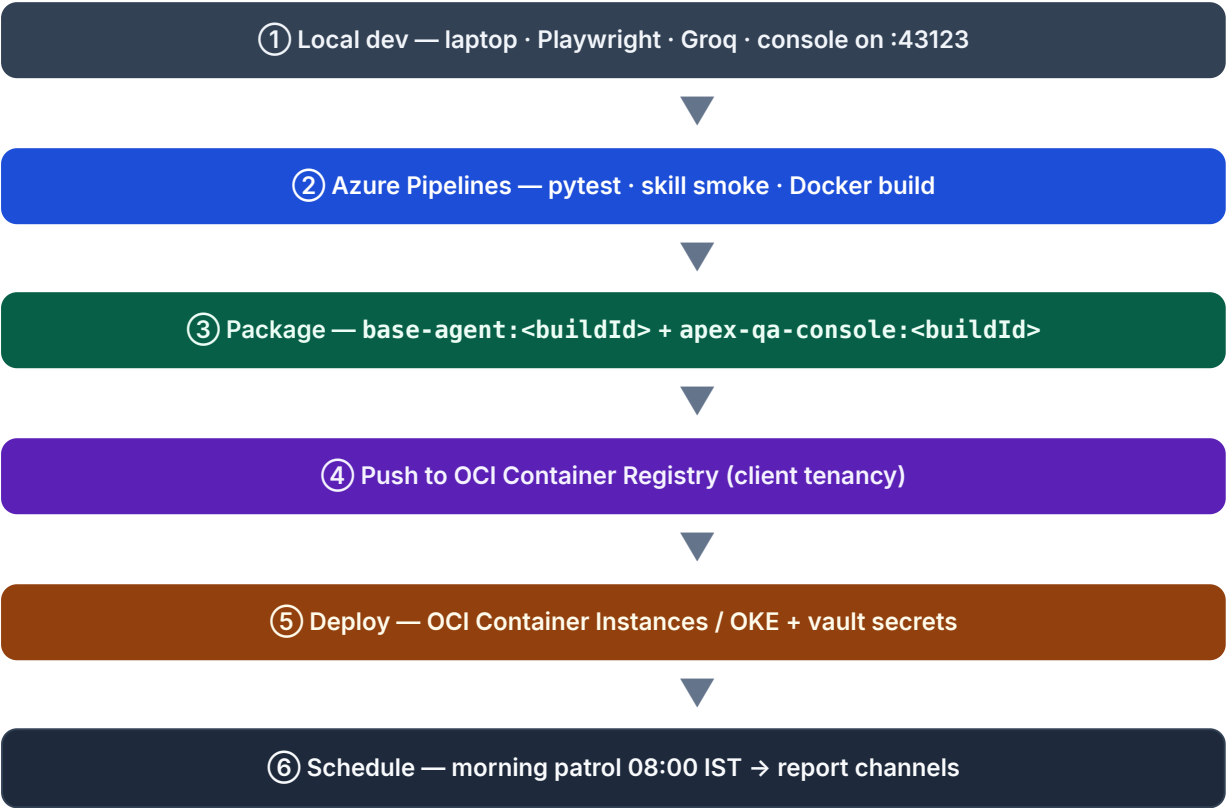
Security Agent (phase 2)

Item	Notes
Scope sign-off	OWASP / APEX subjects
Rules of engagement	Rate limits, hours
Non-prod UAT only	No prod credentials
Shared GT store	tags: [security]

Already built

- Enterprise dark console UI
- KB pack + approval workflow
- Playwright crawler + QA skills
- Groq planner (Claude-ready)
- 26 unit tests passing
- Azure → OCI pipeline skeleton

7. Delivery plan — local → Azure → client OCI



Step	Artifact	Source-code safety
Local	Full repo	Our environment only

CI / Package	Container images	No secrets in git
OCI deploy	Images + vault config	Client never receives raw source

Manual gate: DeployOCI=true in Azure Pipeline after UAT sign-off.

8. Tech stack

Layer	Choice	Why
Runtime	Python 3.10+ · Pydantic	Strong GT/KB contracts; fast iteration
Control plane	LangGraph (bounded)	Explicit state machine — not LLM loops
Browser	Playwright	Reliable APEX SPA automation + evidence
LLM	LiteLLM → Groq / Claude	One gateway; swap models without code change
Console	Next.js 15 · TypeScript · Tailwind	Enterprise dark UI; Vercel/GitHub-inspired UX
CI/CD	Azure Pipelines	Client enterprise standard
Runtime host	OCI containers	Data residency requirement
Secrets	Azure Key Vault / OCI Vault	Credentials never in repo or images

9. Progress summary

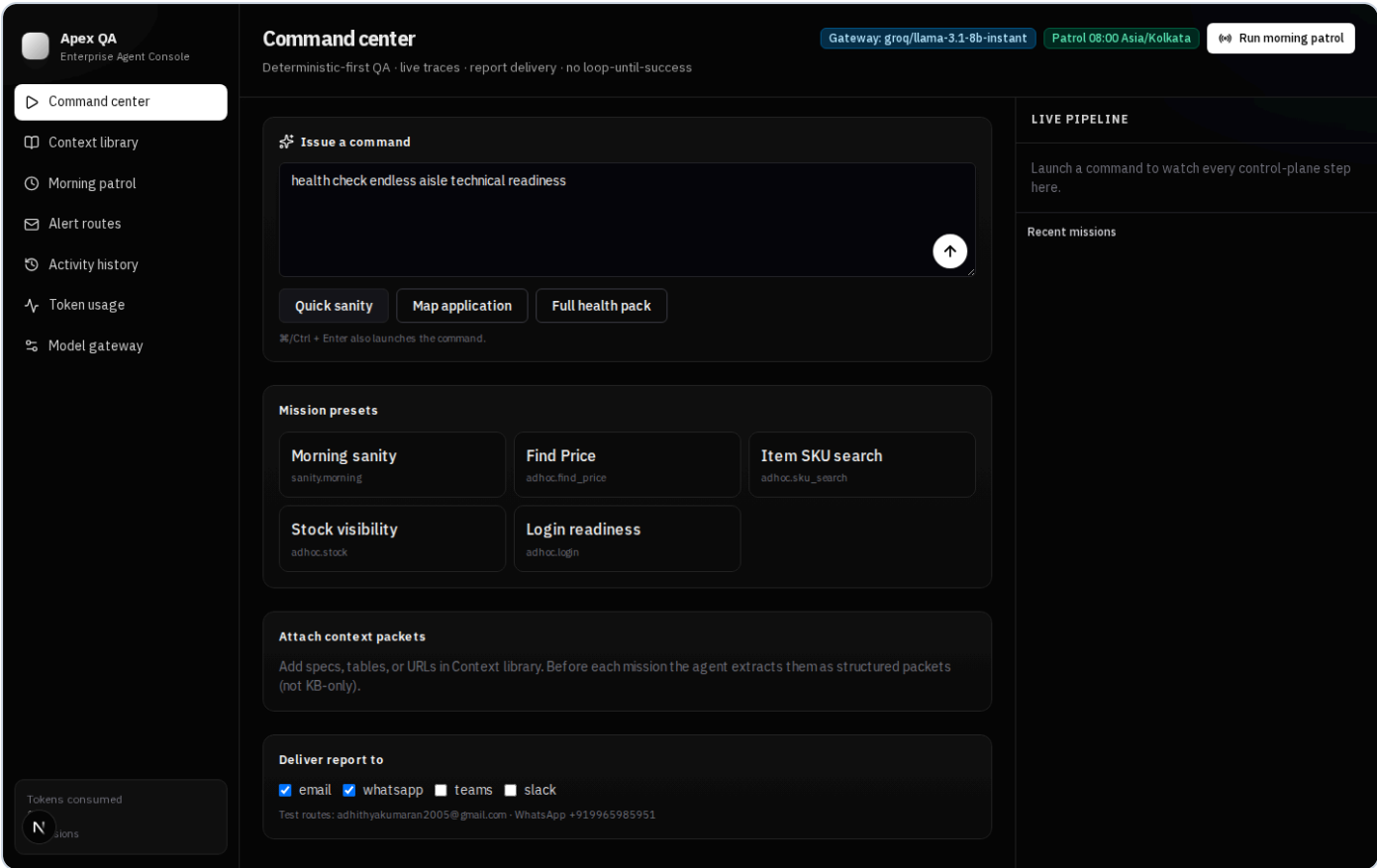
Area	Status
Architecture	Simplified linear path — LLM + Playwright + KB/GT
Console UI	Done — command center, patrol, alerts, model picker
KB / discovery	64 docs from crawl + recordings
Ground Truth	22 candidates — SME session today
LLM planner	Groq wired; Claude-ready
Browser execution	Playwright plugin + local server
Reports	Email / WhatsApp / Teams stubs
Tests	26 unit tests green
Deploy	Azure → OCI pipeline + Dockerfiles

Next after GT today: load approved facts → Phase B validation → live SMTP/Twilio → first OCI deploy.

10. Console UI (screenshots)

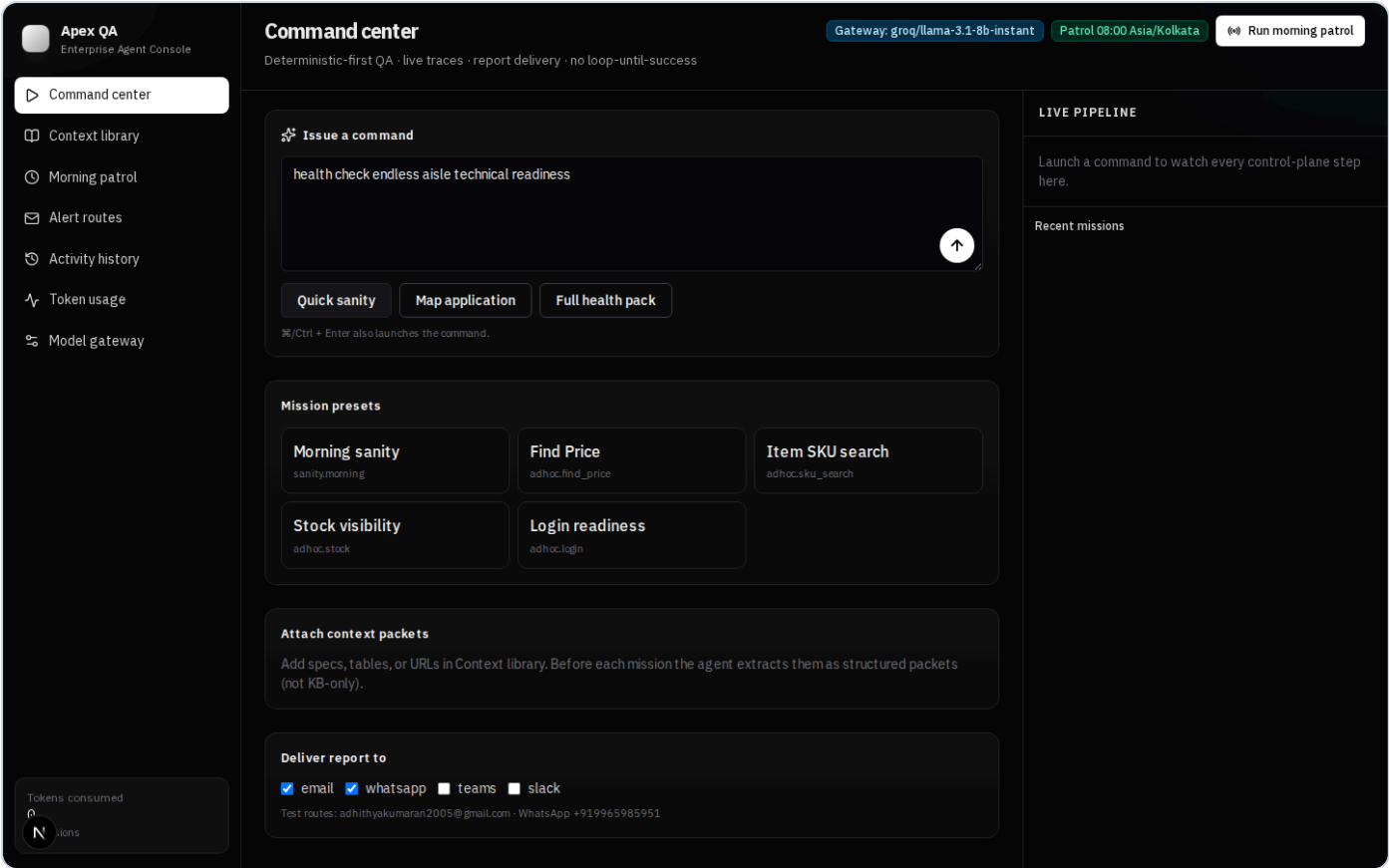
Enterprise dark console — deterministic-first QA with live pipeline, morning patrol, and multi-channel reports.

Figure 1



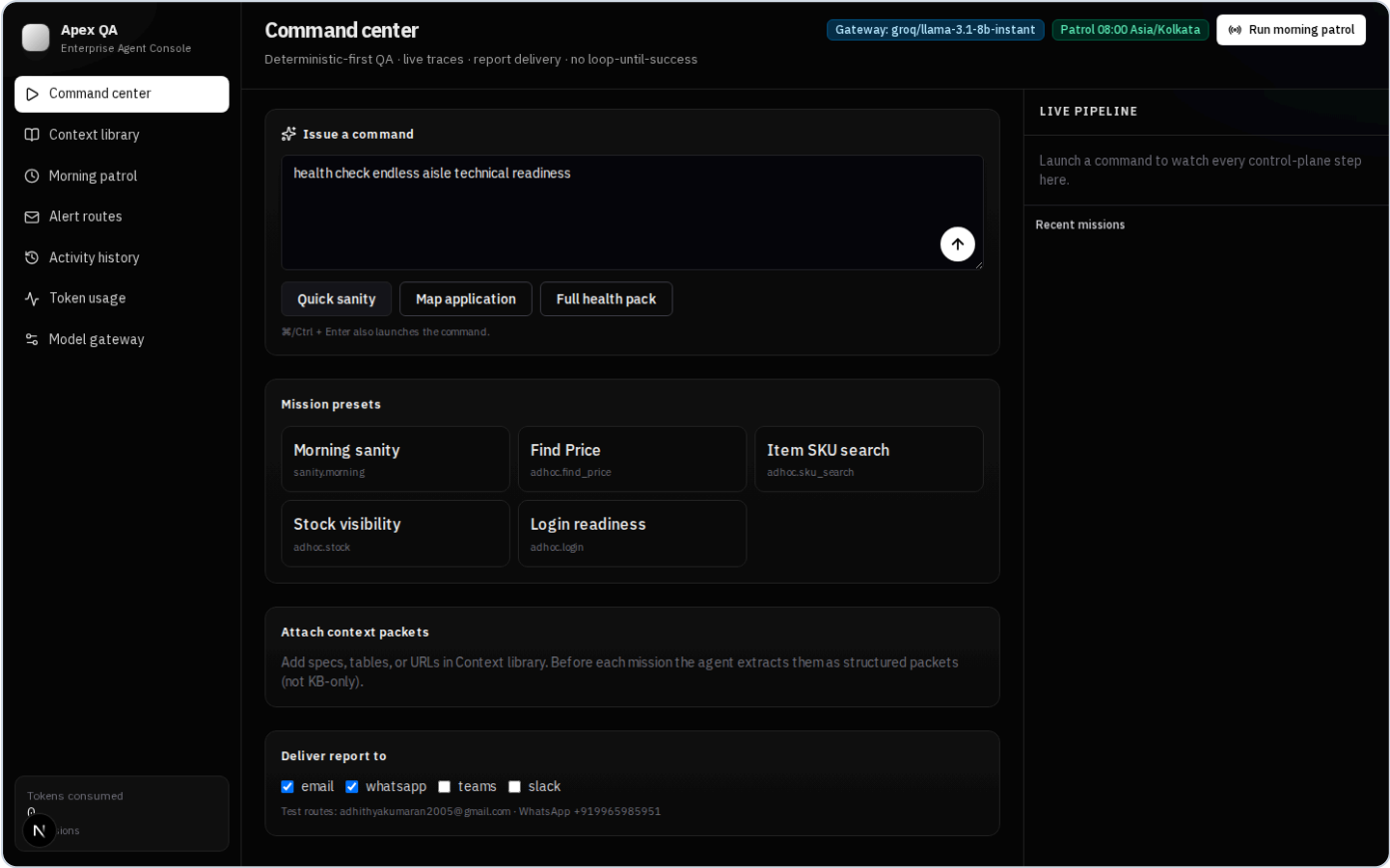
Command center — NL prompt, mission presets, report channels

Figure 2



Mission presets and context packet attachment

Figure 3



Schedule, alert routes, and model gateway

Apex QA Agent Platform · Confidential · Internal / client meeting use